

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO3: Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)

Dave's Taxonomy: Precision (Level 3)

Question Count: 3

Total Marks: 30

Code of Conduct

I J O JOANNA DIVINE (192572003) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : JOANNA

Assessment Tasks

Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS

systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores

product data, transaction logs, and analytics datasets. The system uses platform services for analytics

and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways.

This solution integrates infrastructure, services, and web applications for efficient operations.

Answer:

Case Study 1: Smart Retail Inventory Management System

Introduction

A Smart Retail Inventory Management System is a cloud-based solution designed to help retail stores manage products, stock levels, sales transactions, and demand forecasting efficiently. The system connects Point-of-Sale (POS) systems, store devices, dashboards, and cloud services through secure APIs. Cloud computing provides centralized storage, processing, analytics, scalability, and secure access.

1. System Architecture

The system consists of several layers that work together:

Clients/Store Devices → API Gateway → Cloud Services → Cloud Storage → Analytics & Forecasting

- **Client Layer:** POS systems, computers, tablets, and management dashboards are used by store employees and administrators.

- **API Layer:** APIs allow POS systems and dashboards to communicate with centralized cloud services.
- **Cloud Service Layer:** Cloud-based applications process transactions, update inventory, manage products, and provide business services.
- **Storage Layer:** Cloud storage maintains product information, sales records, transaction logs, and analytics datasets.
- **Analytics Layer:** Platform services analyze historical sales and inventory data to predict future product demand.

2. Working of the System

When a customer purchases a product, the POS system records the transaction. The transaction data is securely sent to the cloud through an API gateway. The cloud application processes the transaction and automatically updates the inventory level in centralized cloud storage.

Managers can access dashboards to view current stock levels, sales information, and inventory reports. The analytics platform analyzes historical sales, seasonal trends, and customer demand to forecast future requirements. Based on these predictions, the retailer can reorder products before stock runs out.

3. Cloud Computing Services Used

The system uses different cloud service models:

Infrastructure as a Service (IaaS)

IaaS provides virtual machines, networking, firewalls, and other computing resources required to run the retail application.

Platform as a Service (PaaS)

PaaS provides application development, database, analytics, and machine-learning services. Demand forecasting can be implemented using cloud-based analytics and machine-learning platforms.

Software as a Service (SaaS)

Web-based dashboards and inventory management applications can be provided as SaaS, allowing managers to access the system through a browser without installing software locally.

4. Data Storage and Management

Cloud storage is used to store:

- Product details and product prices
- Current inventory levels
- Customer purchase transactions
- Sales and transaction logs
- Supplier and order information
- Analytics datasets
- Demand forecasting results

Centralized storage ensures that authorized users and stores can access updated information from different locations.

5. Security Mechanisms

Security is an important part of the system because it handles business and transaction data. The following mechanisms are used:

- **VPN:** Provides secure communication between stores and cloud resources.
- **Firewalls:** Prevent unauthorized network traffic.
- **API Gateway:** Controls and secures API requests between clients and cloud services.
- **Authentication and Authorization:** Ensures that only authorized employees and administrators can access specific resources.
- **Encryption:** Protects sensitive data during transmission and storage.
- **Logging and Monitoring:** Records system activities and helps detect suspicious behavior.

6. Analytics and Demand Forecasting

The cloud platform can analyze historical sales, seasonal patterns, product popularity, and inventory data. Demand forecasting techniques can predict which products are likely to be required in the future.

For example, if the system detects increased demand for umbrellas during the rainy season, the retailer can increase its stock before demand rises. This helps reduce **overstocking, stock shortages, and storage costs**.

7. Benefits of the System

The Smart Retail Inventory Management System provides several benefits:

1. **Real-time inventory visibility** across multiple stores.
2. **Reduced stockouts** through demand prediction.
3. **Lower inventory costs** by avoiding excessive stock.
4. **Centralized data management** using cloud storage.
5. **Scalability** to support additional stores and customers.
6. **Improved decision-making** through analytics and dashboards.
7. **Secure communication** using VPNs, firewalls, and API gateways.
8. **Easy accessibility** through web-based dashboards.

Conclusion

The Smart Retail Inventory Management System demonstrates how cloud computing can integrate infrastructure, platform services, storage, APIs, analytics, and web applications into a single solution. POS systems and dashboards communicate with centralized cloud services through secure APIs, while cloud storage maintains important retail data. Analytics and demand forecasting improve inventory planning, and security mechanisms such as VPNs, firewalls, authentication, encryption, and API gateways protect the system. Therefore, the cloud-based approach provides a **secure, scalable, cost-effective, and efficient solution for modern retail inventory management**.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the

platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed

cloud storage with version control mechanisms. Web APIs enable synchronization across multiple

clients and devices.

Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

Answer:

Case Study 2: Cloud-Based Document Collaboration System

Introduction

A Cloud-Based Document Collaboration System is an online platform similar to Google Docs that allows multiple users to create, edit, share, and manage documents through the Internet. Users can access documents using web browsers from different locations and devices. The system uses cloud computing, distributed storage, web APIs, networking, and security mechanisms to provide real-time collaboration, reliability, and global accessibility.

1. System Architecture

The system can be represented using the following architecture:

Users/Devices → Web Browser → Internet → Cloud Application → Web APIs → Cloud Storage

- **User Layer:** Users access documents through laptops, desktops, tablets, and smartphones.
- **Application Layer:** The cloud application provides document creation, editing, sharing, and collaboration features.
- **API Layer:** Web APIs enable communication and synchronization between different clients and cloud services.
- **Storage Layer:** Documents are stored in distributed cloud storage with version control.

- **Network Layer:** Global cloud networks and data centers provide low latency and high availability.

2. Working of the System

Users first log in to the cloud-based document platform through a web browser. After authentication, they can create a new document or open an existing document.

When multiple users edit the same document, their changes are transmitted to the cloud using web APIs. The system processes these changes and synchronizes them with other connected users in real time. Therefore, users can see updates made by other collaborators almost immediately.

For example, if three employees are working on the same project report from different locations, all three can edit the document simultaneously without maintaining separate copies.

3. Cloud Storage and Version Control

Documents are stored in **distributed cloud storage** rather than on a single local computer. Data can be replicated across multiple storage systems or data centers to improve reliability and availability.

Version control maintains different versions of a document whenever changes are made. It allows users to:

- View previous versions of a document.
- Identify changes made by different users.
- Restore an earlier version when required.
- Prevent accidental loss of important information.
- Maintain a history of document modifications.

This provides better data management and recovery compared with traditional local file storage.

4. Role of Web APIs

Web APIs are responsible for communication between users' devices and cloud services. They allow different clients to send and receive document updates.

APIs support:

- Document synchronization.
- Uploading and downloading files.
- Real-time updates.
- User and permission management.
- Sharing documents between users.
- Communication between different cloud services.

For example, when a user edits a paragraph, the change can be sent through an API to the cloud service, which then synchronizes the updated content with other authorized clients.

5. Network Infrastructure

The system requires a reliable and high-performance network infrastructure because users may access the service from different parts of the world.

Cloud providers use geographically distributed data centers, content delivery mechanisms, load balancing, and high-speed networks to reduce latency and improve availability.

Low latency ensures that changes made by one user are reflected quickly for other users.

High availability ensures that the service continues to operate even if a server or data center experiences a failure.

6. Security Mechanisms

Security is essential because documents may contain confidential enterprise information. The system uses several security mechanisms:

Encryption

Data is encrypted during transmission and while stored in cloud storage. This protects documents from unauthorized access.

Identity Management

Identity management verifies the identity of users through authentication mechanisms such as usernames, passwords, and multi-factor authentication (MFA).

Access Permissions

Administrators can define who can view, edit, comment on, or share a document. For example, an employee may be given editing permission while another user may have view-only permission.

Auditing and Monitoring

User activities can be logged and monitored to identify unauthorized access or suspicious activities.

7. Benefits of the System

The Cloud-Based Document Collaboration System provides several advantages:

1. **Real-time collaboration** between multiple users.
2. **Global accessibility** from different locations and devices.
3. **Centralized document storage** in the cloud.
4. **Automatic synchronization** across multiple clients.
5. **Version control** for tracking and recovering document changes.
6. **High availability** through distributed cloud infrastructure.
7. **Reduced infrastructure cost** because organizations do not need to maintain large local storage systems.
8. **Improved security** through encryption, identity management, and access permissions.
9. **Scalability** to support increasing numbers of users and documents.

Conclusion

The Cloud-Based Document Collaboration System demonstrates how cloud computing can provide efficient and secure online collaboration. Users can access and edit documents through browsers while web APIs synchronize changes across multiple devices. Distributed cloud storage provides scalability, reliability, and version control, while global network infrastructure provides low latency and high availability. Security mechanisms such as encryption, identity management, authentication, and access permissions protect enterprise documents. Therefore, this system provides a **scalable, reliable, secure, and cost-effective solution for real-time document collaboration.**

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients.

Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.

Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure

uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

through authentication tokens, HTTPS, and role-based access control. The system demonstrates

integration of clients, APIs, storage, and platform services in real time.

Answer:

Case Study 3: Online Food Ordering Platform

Introduction

An Online Food Ordering Platform is a cloud-based application that allows customers to browse restaurants and menus, place food orders, make payments, and track their orders using web browsers or mobile applications. The system uses cloud computing to connect frontend clients with backend services through RESTful Web APIs. Cloud storage is used to securely store customer information, menu images, invoices, and other application data. Virtual machines and managed Kubernetes provide scalability, reliability, and efficient resource management.

1. System Architecture

The basic architecture of the online food ordering platform is:

Web/Mobile Clients → RESTful Web APIs → Cloud Backend Services → Cloud Storage

The major components are:

- **Frontend Clients:** Web browsers and mobile applications through which customers interact with the system.
- **RESTful Web APIs:** Provide communication between frontend applications and backend services.
- **Backend Services:** Process user registration, restaurant details, menu information, orders, payments, and order tracking.
- **Cloud Storage:** Stores customer data, menu images, invoices, and other application files.
- **Virtual Machines:** Provide computing resources for running backend applications.
- **Managed Kubernetes:** Manages containers and automatically scales application services based on demand.

2. Working of the System

First, the customer accesses the platform using a web browser or mobile application. The customer can register or log in, browse available restaurants and menus, select food items, and place an order.

The frontend sends requests to the backend through RESTful Web APIs. The backend processes the request and communicates with databases and cloud storage. After an order is placed, the system stores the order information and generates an invoice.

The restaurant receives the order through the platform and updates its status, such as **Order Accepted, Food Preparing, Ready for Delivery, and Delivered**. These updates are sent through cloud services and displayed to the customer in real time.

3. Role of RESTful Web APIs

RESTful Web APIs act as a communication layer between frontend clients and backend cloud services.

They can provide operations such as:

- User registration and login.
- Restaurant and menu retrieval.

- Adding food items to the cart.
- Placing and updating orders.
- Payment processing.
- Order tracking.
- Invoice generation.
- Customer and restaurant management.

For example, when a customer selects a food item, the mobile application sends a request to the appropriate API. The backend processes the request and returns the required information to the application.

4. Cloud Storage

Cloud storage is used to securely store different types of information, including:

- Customer details.
- Restaurant and menu information.
- Food and menu images.
- Order information.
- Digital invoices.
- Application files and logs.

Cloud storage provides scalability, backup, reliability, and easy access to data for authorized cloud services.

5. Virtual Machines and Kubernetes

Virtual Machines

Virtual machines provide computing resources for running the application's backend services. They can be configured according to the processing requirements of the application.

Managed Kubernetes

Managed Kubernetes is used to deploy and manage containerized application services. It provides:

- Automatic scaling of application containers.
- Load distribution between services.
- Automatic restarting of failed containers.
- Efficient resource utilization.
- Easy deployment and management of application updates.

For example, during lunch or dinner hours, the number of food orders may increase significantly. Kubernetes can increase the number of running application instances to handle the additional workload.

6. Security Mechanisms

Security is important because the platform handles customer accounts, orders, and other sensitive information.

Authentication Tokens

Authentication tokens are issued after successful login and are used to verify requests made by authenticated users.

HTTPS

HTTPS encrypts communication between clients and cloud services. It helps protect customer information and API requests from unauthorized interception.

Role-Based Access Control (RBAC)

RBAC provides different permissions based on the user's role. For example:

- **Customer:** Browse menus, place orders, and view order status.
- **Restaurant Staff:** Manage menus and update order status.
- **Administrator:** Manage users, restaurants, orders, and system settings.

This prevents users from accessing functions that are not authorized for their roles.

7. Scalability and Reliability

The cloud-based architecture allows the system to handle changes in user traffic. Virtual machines provide computing resources, while Kubernetes can automatically scale application containers according to workload.

Load balancing distributes requests across multiple application instances. If one instance fails, another instance can continue serving users. This improves the availability and reliability of the platform.

8. Real-Time Integration

The system integrates frontend clients, APIs, storage, and cloud platform services in real time.

For example:

Customer places order → API receives request → Backend processes order → Order stored in cloud → Restaurant receives order → Restaurant updates status → Customer sees updated status

This integration provides a fast and seamless ordering experience.

9. Benefits of the System

The cloud-based food ordering platform provides several benefits:

1. **Easy accessibility** through web browsers and mobile applications.
2. **Real-time order processing and tracking.**
3. **Scalability** during peak ordering periods.
4. **High availability and reliability** using cloud infrastructure and Kubernetes.
5. **Secure data storage** for customer information and invoices.
6. **Efficient communication** through RESTful Web APIs.
7. **Flexible resource management** using virtual machines and containers.
8. **Role-based security** to control access to different functions.
9. **Reduced infrastructure maintenance** because cloud services manage much of the underlying infrastructure.

Conclusion

The Online Food Ordering Platform is a good example of cloud-based application architecture. Web and mobile clients communicate with backend services through RESTful Web APIs, while cloud storage securely maintains customer data, menu images, and invoices. Virtual machines provide computing resources, and managed Kubernetes enables scalability, fault tolerance, and reliable service delivery. Authentication tokens, HTTPS, and role-based access control protect the application and its data. Thus, the system provides a secure, scalable, reliable, and real-time platform for online food ordering and delivery management.